

TROLL A OIL PLATFORM, Norway

WMO: 010877

Lat: 60.633N

Lon: 3.717E

Elev: 90

StdP: 100.25

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.0	0.0	-10.8	1.5	0.7	-8.1	1.9	1.6	32.4	2.8	30.4	3.2	11.2	150	1.128

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	20.2	15.4	18.1	14.3	16.9	14.1	16.3	19.2	15.4	17.4	14.7	16.3	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.9	16.6	14.8	10.6	16.3	14.0	10.1	15.6	45.8	18.9	43.3	17.2	41.5	15.9	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.4	24.4	22.1	DB	-2.6	24.9	2.7	2.7	-4.5	26.8	-6.1	28.4	-7.6	29.9	-9.5	31.9	(4)
(5)			WB	-4.5	17.4	2.5	1.8	-6.3	18.7	-7.8	19.7	-9.3	20.7	-11.1	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	4.5	4.2	4.7	6.4	8.8	11.1	13.7	14.3	12.4	9.8	7.4	5.8		
	DBStd	4.08	2.01	2.04	2.13	2.32	2.71	1.86	2.65	1.67	1.87	1.82	2.09	2.13		
	HDD10.0	904	172	161	163	111	55	8	0	0	2	23	79	129		
	HDD18.3	3548	430	394	422	357	296	216	151	127	178	263	327	387		
	CDD10.0	403	0	0	0	3	18	42	113	132	74	18	2	0		
	CDD18.3	7	0	0	0	0	1	0	6	1	0	0	0	0		
	CDH23.3	13	0	0	0	0	1	0	12	0	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	11.1	13.3	13.5	11.1	9.7	9.9	9.4	8.8	9.2	10.3	12.0	12.0	14.1		
(15) Precipitation	PrecAvg	1857	203	170	158	110	83	98	111	141	165	197	198	223		
	PrecMax	2451	364	342	321	214	130	220	174	253	289	357	367	398		
	PrecMin	1315	37	16	34	28	20	12	26	55	56	61	46	55		
	PrecStd	286	86	78	69	38	28	45	36	46	73	69	84	88		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	8.2	9.2	15.0	20.1	18.2	25.1	20.9	18.2	14.8	12.2	10.2		
		MCWB	8.7	7.6	7.4	8.5	13.6	12.7	16.9	14.6	13.9	11.3	10.9	9.6		
	2%	DB	8.2	8.0	8.2	12.2	16.9	15.9	21.9	18.9	16.8	13.2	11.2	9.8		
		MCWB	7.7	7.4	7.3	7.7	11.9	12.7	16.6	14.6	13.3	11.2	9.8	9.3		
	5%	DB	8.0	7.2	8.0	10.2	14.2	14.1	19.1	17.2	15.9	12.8	10.9	9.1		
		MCWB	7.5	6.6	7.2	7.1	10.2	12.1	15.5	14.6	13.2	10.9	9.5	8.5		
	10%	DB	7.8	7.0	7.2	9.0	12.1	13.1	16.9	16.2	14.9	12.1	10.1	8.8		
		MCWB	7.1	6.4	6.3	7.2	9.9	11.5	14.8	14.4	13.0	10.3	9.0	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.2	8.4	9.5	14.2	14.2	18.7	16.7	15.1	13.2	11.5	10.0		
		MCDB	8.7	8.2	8.5	11.3	19.6	16.4	21.5	18.3	16.7	13.7	12.1	10.2		
	2%	WB	8.1	7.3	7.9	8.6	12.4	13.2	17.0	15.7	14.2	12.2	10.8	9.2		
		MCDB	8.2	7.5	8.0	10.2	15.8	14.8	21.5	17.3	15.6	12.8	11.4	9.3		
	5%	WB	7.6	6.7	7.2	8.1	11.1	12.6	15.9	15.3	13.8	11.6	10.1	8.6		
		MCDB	7.7	7.0	7.4	9.2	13.3	14.0	18.9	16.7	15.2	12.3	10.5	8.8		
	10%	WB	6.8	6.3	6.6	7.5	10.1	11.9	15.0	14.8	13.4	10.9	9.4	8.0		
		MCDB	7.2	6.7	7.0	8.5	11.5	13.2	16.6	16.1	14.7	11.9	9.8	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.7	2.7	2.8	3.0	3.3	2.7	3.0	2.8	2.5	2.4	2.4	2.8		
		MCDBR	2.9	2.9	3.0	4.9	7.2	4.9	6.8	4.3	3.8	3.0	2.5	2.9		
	5% WB	MCWBR	3.4	3.3	2.9	2.7	3.3	2.9	3.2	2.8	2.4	2.8	2.8	3.5		
		MCWBR	2.8	2.7	2.6	3.5	6.2	4.2	5.6	3.6	3.1	2.8	2.4	2.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.343	0.358	0.378	0.364	0.359	0.375	0.371	0.361	0.340	0.312	0.398		
		taud	2.103	2.237	2.311	2.311	2.427	2.476	2.452	2.459	2.449	2.349	2.169	2.443		
	Ebn at Noon	Ebn at Noon	394	628	759	815	862	872	845	819	758	631	401	262		
		Edh at Noon	44	70	89	106	101	98	99	91	78	61	41	30		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.21	0.73	1.87	3.63	5.12	5.30	4.79	3.70	2.25	1.07	0.33	0.13		
	RadStd	RadStd	0.03	0.12	0.26	0.37	0.37	0.57	0.38	0.31	0.20	0.14	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (31 neighbors)		N/A	+0.65	N/A	N/A	N/A	N/A	-83	-118	N/A	N/A	(47)

Nomenclature: See separate page